

12 **(long)** The area of the two plates of a parallel plate capacitor is  $A$ , and the distance between them is  $d$ . There is a parallel dielectric plate of the same area with a thickness of  $t$  and a dielectric constant of  $\epsilon_r$  between them, as shown in Fig. 7. Initially, switch  $S$  is ON, which charges the capacitor to a voltage of  $V$ . Treat the problem as capacitors in series and ignore the fringe effect.

- (a) Calculate the capacitance of the capacitor with the dielectric plate. [6]
- (b) Disconnect the power supply by switching off the switch  $S$  and pull out the dielectric plate. How much work needs to be done? [12]
- (c) If the dielectric plate is pulled out without disconnecting the power supply, how much work will be done? [12]

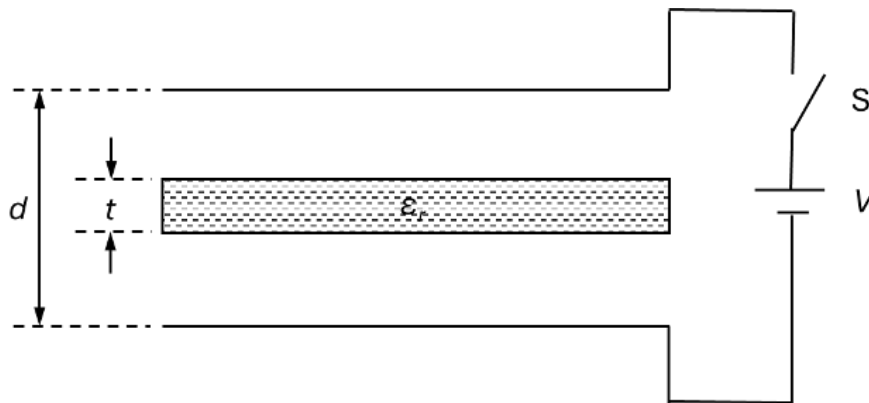


Fig. 7

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